Gene Expression Analysis Across Pregnancy Samples

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1 Methods

1.1 Data

I used gene expression data from 157 pregnancy samples in the GSE85307 dataset. The data included maternal age, BMI, gestational age at delivery, and pregnancy status.

There were 110 normal pregnancies and 47 pregnancies with preeclampsia.

I focused on three genes: *CNR1*, *SOCS5*, and *SIRT3*.

1.2 Data Preparation

I used R to match the gene expression data with maternal information using sample IDs.

I matched gene names with probe IDs using the gene lookup file.

I used maternal BMI as a continuous variable and pregnancy status as a categorical variable.

I also divided BMI into two groups, below 25 and 25 or higher.

1.3 Analysis

I made a histogram, scatterplot, and boxplot for each of the three genes, giving nine plots.

I calculated summary statistics and Pearson correlations with maternal BMI.

I also made a table of maternal characteristics by pregnancy status, a heatmap of ten genes with clustering and two annotation bars, and a parallel coordinates plot for the three selected genes.

I used R 4.6.1, RStudio, and R Markdown for the analysis.

2 Results

2.1 Maternal Characteristics

The dataset had 157 samples, including 110 normal pregnancies and 47 cases of preeclampsia.

Table 1: Maternal characteristics by pregnancy status

Characteristic	Overall N=157	Normal N=110	Preeclampsia N=47
Maternal age (years)	26.38 (5.04)	26.63 (5.08)	25.78 (4.95)
Maternal BMI	28.94 (7.88)	27.99 (7.66)	31.16 (8.03)
Gestational age (weeks)	38.82 (2.18)	39.25 (1.02)	37.81 (3.49)
BMI < 25	53 (34%)	40 (36%)	13 (28%)
BMI ≥ 25	104 (66%)	70 (64%)	34 (72%)

Continuous values are mean (SD). BMI categories are n (%).

The preeclampsia group had a higher average BMI (31.16) than the normal group (27.99). Average gestational age at delivery was lower in the preeclampsia group (37.81 versus 39.25 weeks). These are descriptive differences.

2.2 Gene Expression Summary

Table 2: Descriptive statistics for the three genes

Gene	Mean	Median	SD	Minimum	Maximum
CNR1	4.873	4.861	0.140	4.549	5.574
SOCS5	7.474	7.479	0.146	7.123	7.888
SIRT3	8.916	8.923	0.123	8.565	9.290

SIRT3 had the highest mean expression, and *SOCS5* had the largest standard deviation. These values describe the three expression distributions in this dataset.

2.3 Maternal BMI and Gene Expression

Table 3: Pearson correlations between BMI and gene expression

Gene	Correlation
CNR1	-0.093
SOCS5	0.084
SIRT3	-0.039

All three correlations were weak. I did not see much of a linear relationship between BMI and expression of these genes.

2.4 Gene Expression Plots

The histograms show the expression distributions. The scatterplots compare expression with BMI, and the boxplots compare pregnancy status and BMI groups.

The boxplots show overlapping values across the groups.

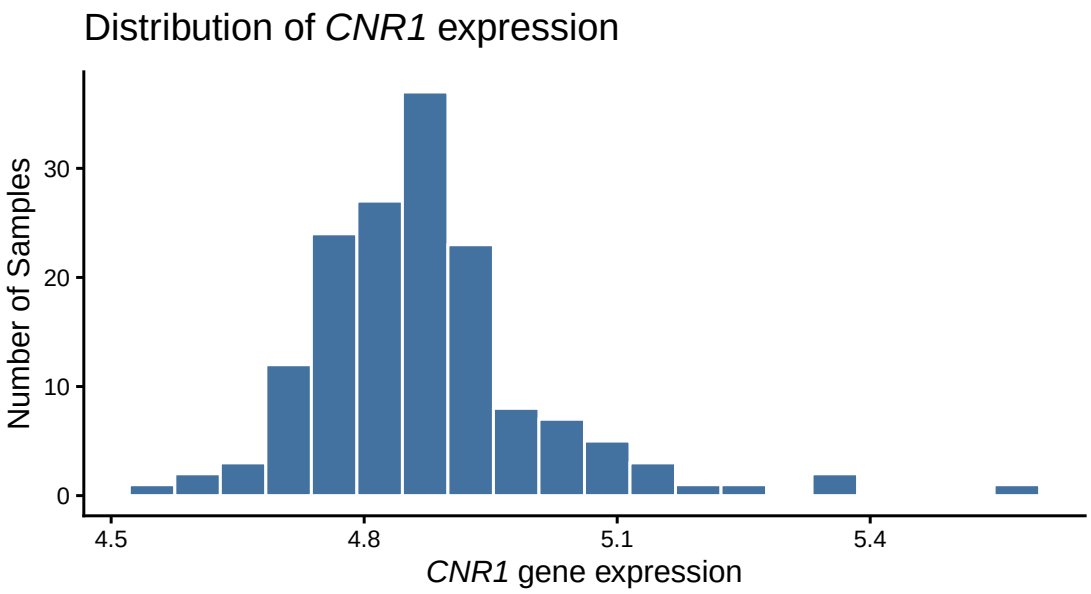


Figure 1: Distribution of *CNR1* expression.

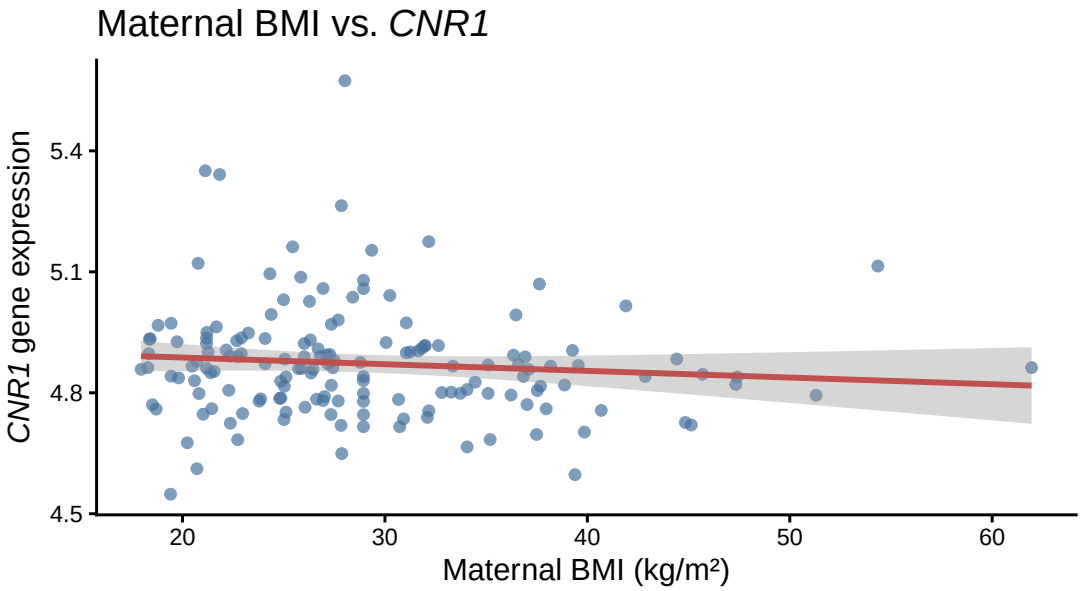


Figure 2: Maternal BMI and *CNR1* expression.

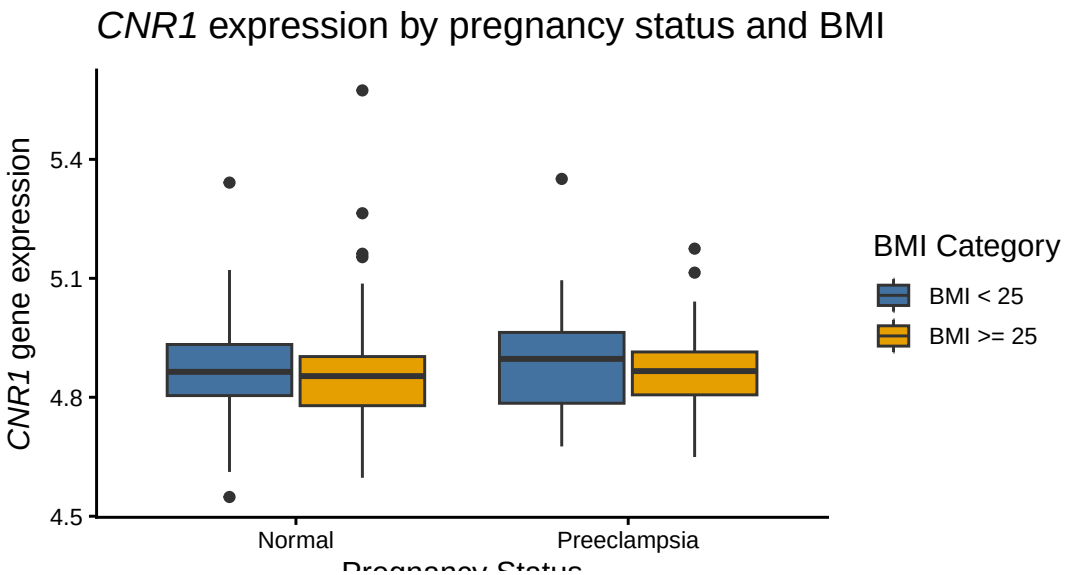


Figure 3: *CNR1* expression by pregnancy status and BMI group.

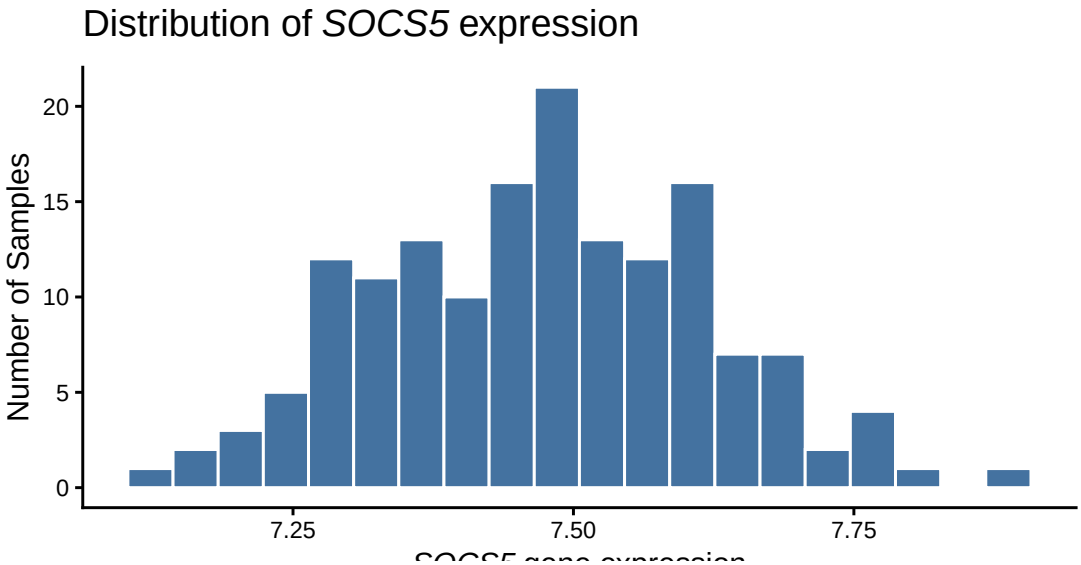


Figure 4: Distribution of *SOCS5* expression.

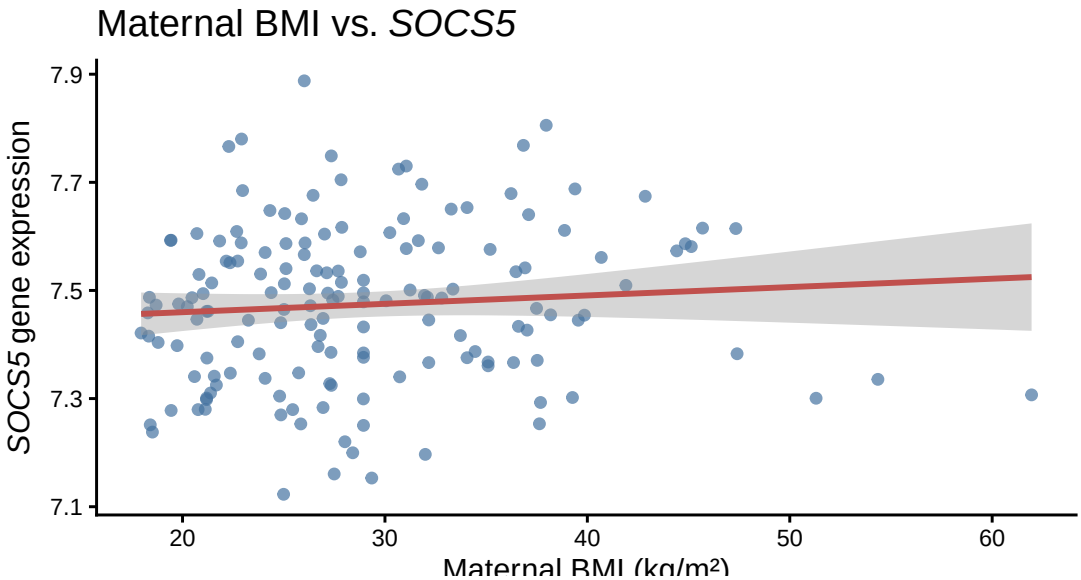


Figure 5: Maternal BMI and *SOCS5* expression.

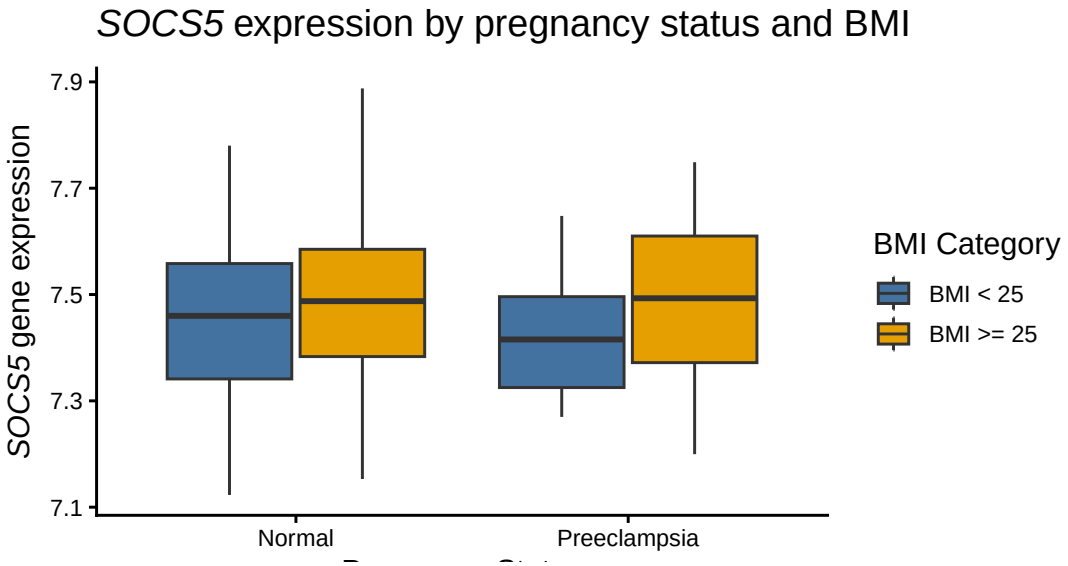


Figure 6: *SOCS5* expression by pregnancy status and BMI group.

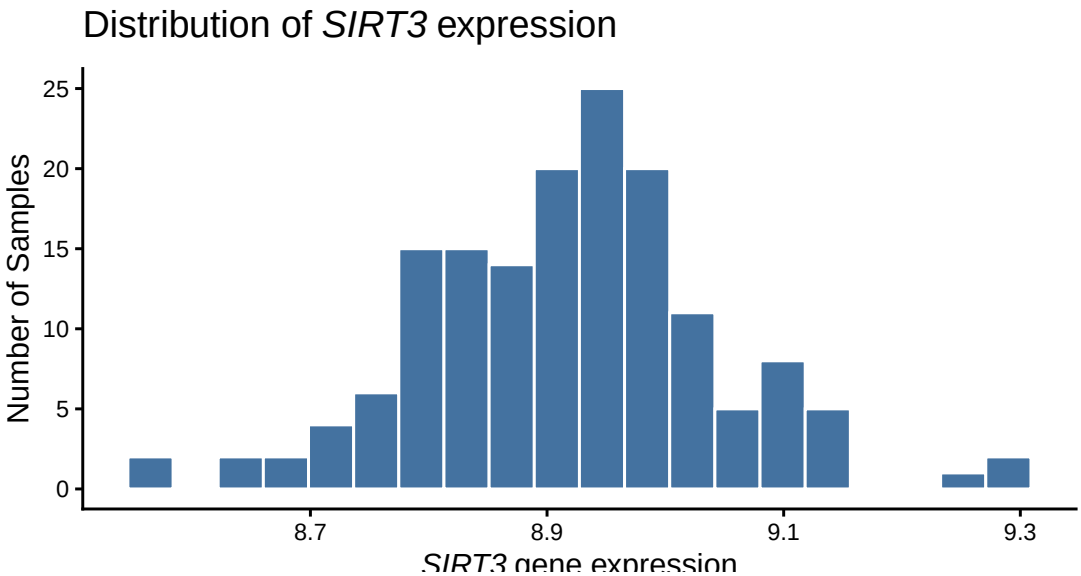


Figure 7: Distribution of *SIRT3* expression.

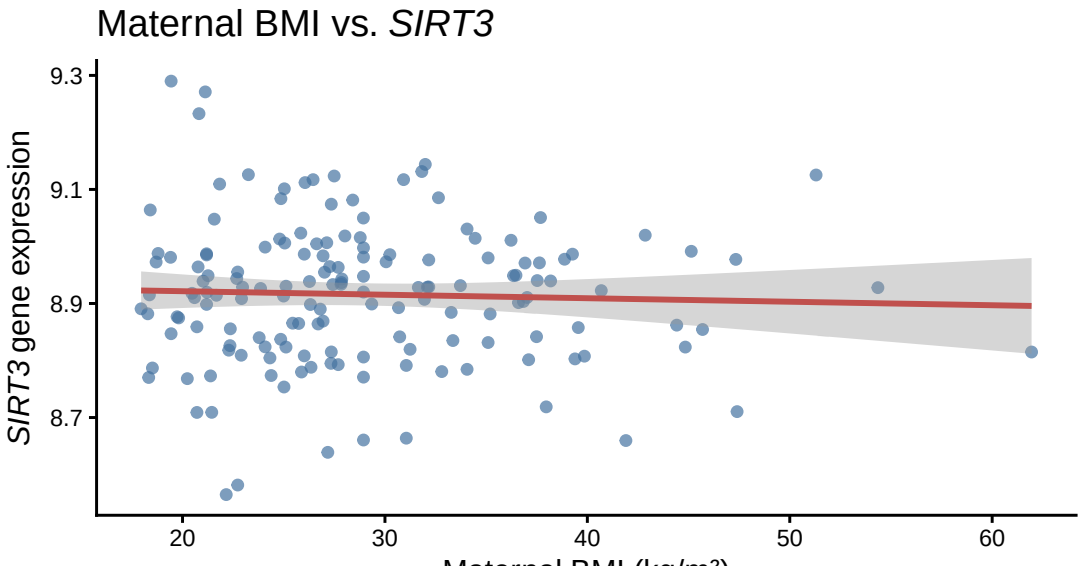


Figure 8: Maternal BMI and *SIRT3* expression.

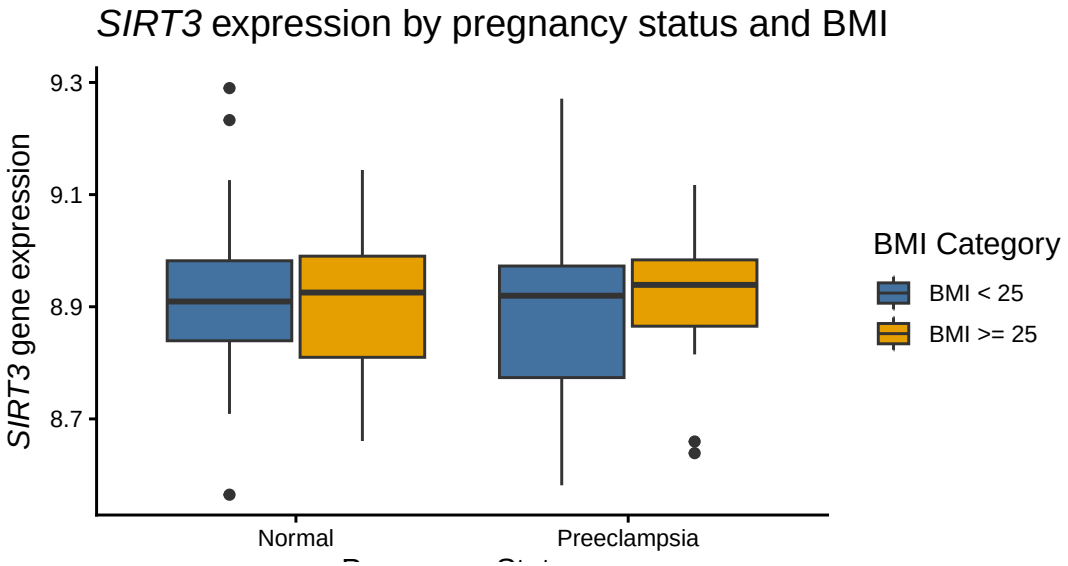


Figure 9: *SIRT3* expression by pregnancy status and BMI group.

2.5 Heatmap

I made a heatmap of ten genes across the 157 samples.

I standardized values within each gene, used hierarchical clustering, and added pregnancy status and BMI group as annotation bars.

I did not see a clear separation between the pregnancy groups from this plot alone.

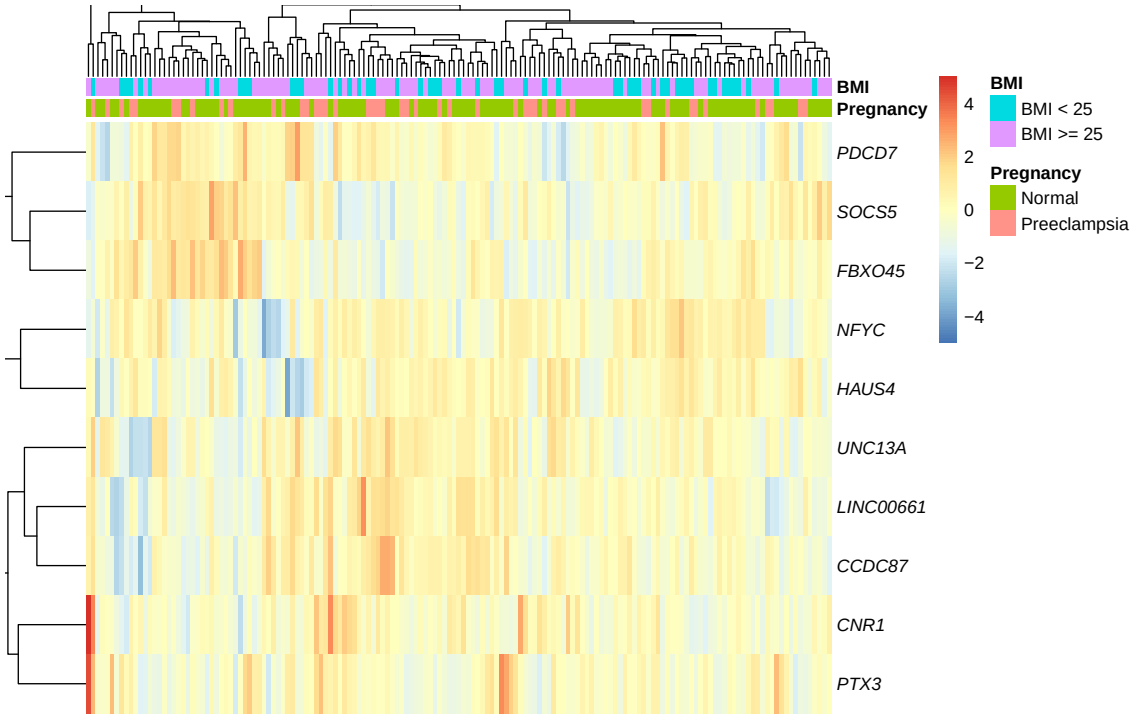


Figure 10: Heatmap of ten genes with clustering and two annotation bars.

2.6 Parallel Coordinates Plot

I used a parallel coordinates plot to compare standardized expression of *CNR1*, *SOCS5*, and *SIRT3*.

Thin lines show individual samples, and thicker lines show the group averages.

The average profiles looked similar between normal pregnancies and preeclampsia.

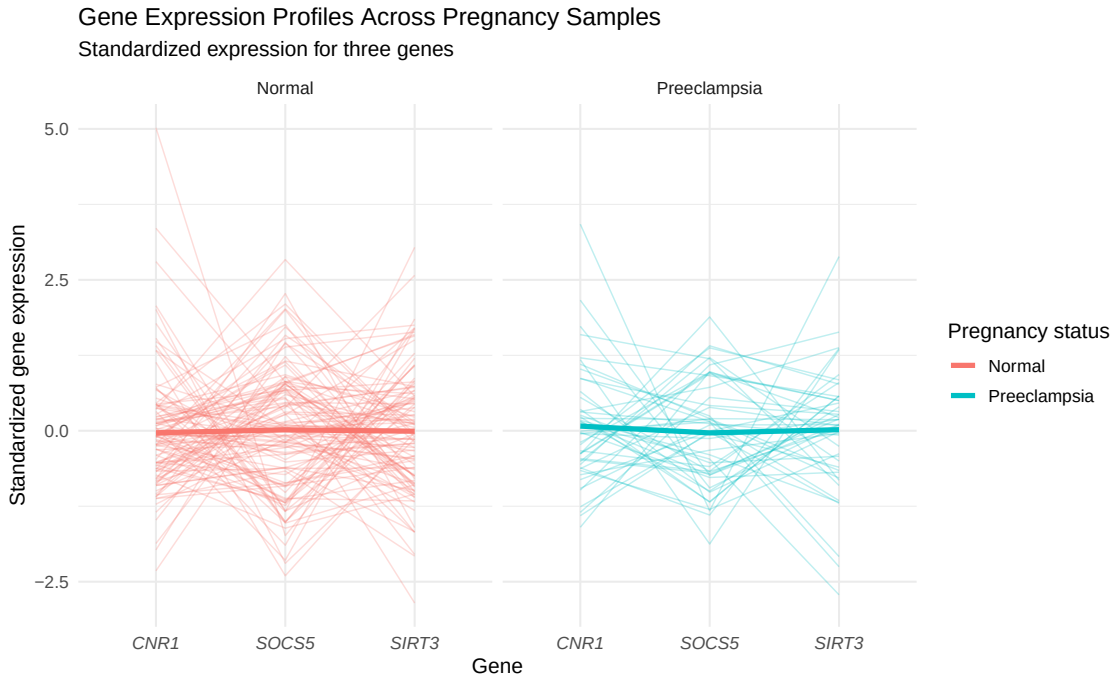


Figure 11: Parallel coordinates plot of three genes by pregnancy status.

3 References

R Core Team. *R: A Language and Environment for Statistical Computing*. `https://www.r-project.org/`.

Wickham, H. *ggplot2: Elegant Graphics for Data Analysis*. `https://ggplot2.tidyverse.org/`.

NCBI Gene Expression Omnibus. Dataset GSE85307. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE85307`.